nvfp4_dual_gemm

Ended 2026-01-20 07:59 UTC

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▼ Description

You will implement a block scaled dual matrix-matrix multiplication kernel with silu activation optimized for NVIDIA B200. To be explicit, you will be given a tuple of tensors:

```
(a, b1, b2, sfa, sfb1, sfb2, c)
```

where:

- `a` is M x K x L in K-major order in nvfp4(e2m1)
- `b1` is N x K x L in K-major order in nvfp4(e2m1)
- `b2` is N x K x L in K-major order in nvfp4(e2m1)
- `sfa` is M x (K // 16) x L in K-major order in fp8(e4m3fnuz)
- `sfb1` is N x (K // 16) x L in K-major order in fp8(e4m3fnuz)
- `sfb2` is N x (K // 16) x L in K-major order in fp8(e4m3fnuz)
- `c` is M x N x L in fp16

Matrix sizes `M` is divisible by mma_tiler_mn[0], `N` is divisible by mma_tiler_mn[1], `K` is divisible by 256. The ranking criteria is the geometric mean of the benchmark results. For the grand price, your kernel will be evaluated against the speed of light analysis and the solution closest to the speed of light will be awarded the grand price.

```
The speed of light analysis based on the max(FP4 Tensor Core math throughput, D
  M   N   K   L time[us]
256 4096 7168 1 4.708
512 4096 7168 1 8.714
256 3072 4096 1 2.125
512 3072 7168 1 6.535
```

▼ Benchmark Shapes

- `{"k":7168,"l":1,"m":256,"n":4096}`
- `{"k":7168,"l":1,"m":512,"n":4096}`
- `{"k":4096,"l":1,"m":256,"n":3072}`

- {"k":7168,"l":1,"m":512,"n":3072}

RANKINGS REFERENCE SUBMISSION

NVIDIA

Hide

guaguabear	12.913µs		try.py
gau.nernst	13.124µs	+0.210µs	submission_v2.py
arseni_ivanov	13.876µs	+0.752µs	temp.py
Simon	14.055µs	+0.179µs	submission.py
macto	14.146µs	+0.091µs	submission.py
rex_cz	14.164µs	+0.018µs	nvfp4_dual_gemm_v6.py
Arseni Ivanov	14.172µs	+0.008µs	umma_based_working.py
jiab_85281	14.174µs	+0.002µs	sub_v3.py
crUst	14.176µs	+0.003µs	submission.py
basesearch	14.178µs	+0.002µs	submission.py
shigao	14.181µs	+0.003µs	submission.py
shiyegao	14.197µs	+0.015µs	submission.py
novo_force	14.253µs	+0.056µs	submission.py
gum	14.522µs	+0.269µs	wc5.py
rt11	14.561µs	+0.039µs	v_ai_8_1.py
leymore4172	14.619µs	+0.059µs	submission.py
LemonAndRabbit	14.642µs	+0.022µs	submission_11.py
hekailove	14.704µs	+0.063µs	result.py
shiyeegao	14.769µs	+0.065µs	node_104.py
mufeez-amjad	14.833µs	+0.064µs	v6.py
snowclipsed	14.834µs	+0.000µs	solution.py
LemonStar	14.838µs	+0.004µs	submission_10_2.py

XoTic	14.979µs	+0.141µs	v6.py
Quantizr	15.119µs	+0.140µs	yuh.py
Naturalseeker	15.173µs	+0.053µs	solution_umxpppp.py
yue	15.226µs	+0.053µs	optimize_try0.py
nrehiew	15.244µs	+0.018µs	submission.py
binsquare	15.812µs	+0.568µs	submission.py
swanbomb_	16.047µs	+0.235µs	tvm.py
pongtsu	16.168µs	+0.121µs	v2.7.py
	16.347µs	+0.179µs	submission.py
gearfive	16.363µs	+0.016µs	submission_cuda.py
jihun_lee__	16.765µs	+0.402µs	submission.py
rootacess	17.404µs	+0.640µs	submission1e.py
MINJAE	18.264µs	+0.860µs	submission_v18_2.py
apemon.eth	18.267µs	+0.003µs	kernel.py
jason	18.426µs	+0.159µs	submission_v7.py
_spatters	20.274µs	+1.848µs	v1.py
currybab	20.413µs	+0.139µs	submission_4.py
Sambhav	20.490µs	+0.077µs	nvfp4_dual_gemm_v2.py
whoknows	20.503µs	+0.013µs	prob3_cutedsl.py
dxyz	20.512µs	+0.009µs	submission_v0.py
shikhar	20.555µs	+0.042µs	nvfp4_dual_gemm_v2.py
Kazim	20.756µs	+0.202µs	nvfp4_dual_gemmK4.py
Will	21.318µs	+0.561µs	cute_submission_716.py
vyom	22.366µs	+1.048µs	submission.py
ozamatash	22.895µs	+0.529µs	submission.py
HayatoFujihara	23.603µs	+0.708µs	nvfp4_dual_gemm_v12.py
whatdhack_	23.659µs	+0.056µs	dual_gemm_opt_cute.py
VladRad	24.473µs	+0.814µs	submission.py
Gusarich	29.583µs	+5.110µs	submission.py
Phạm Lê Huy Hoàng 2	30.309µs	+0.726µs	submission.py

Batuhanaktas	30.479µs	+0.170µs	kai_optimizer.py
lucifer_0000007	30.633µs	+0.153µs	submissio.py
Clark Kitchen	31.332µs	+0.699µs	submission_best_v6_1.py
baicai1145	32.462µs	+1.130µs	submission.py
houdinied	33.790µs	+1.328µs	v1sub.py
phuc9702	36.253µs	+2.463µs	nvfp4_dual_gemm.py
dandanaka_hitman	38.262µs	+2.009µs	mrk2.py
Joel🏴	39.101µs	+0.839µs	submission-v7.py
Alessandro	39.246µs	+0.145µs	submission.py
jethreetwo	39.681µs	+0.435µs	v6.py
vesper15	41.019µs	+1.338µs	submission2.py
prashantgoswami038	41.621µs	+0.602µs	nvfp4_dual_gemm.py
francescagreco	43.110µs	+1.489µs	chew2ULTRAsolgems.py
zyvren	43.462µs	+0.352µs	optimized_submission.py
Carlos Andrés Chimae	47.320µs	+3.858µs	nvfp4_dual_gemm.py
symlon	56.132µs	+8.812µs	inline.py
nikitka93	56.183µs	+0.051µs	nvfp4_v19_fixed_Jan16.py
ago.lajko	56.937µs	+0.754µs	dual_triton_sub.py
drillyb	57.224µs	+0.287µs	opyimized.py
nwyin	57.242µs	+0.018µs	nvfp4_dual_gemm.py
wkwkes9611	57.548µs	+0.306µs	opt.py
boo	60.446µs	+2.898µs	reference7_fast_v22.py
Ram	63.210µs	+2.764µs	submission_3stage.py
pico	66.920µs	+3.710µs	submission_v1.py
rl5409_44612	69.630µs	+2.709µs	nvfp4_dual_gemm.py
ra_XOr	69.659µs	+0.029µs	submission_-_ref_1.py
Samy	69.665µs	+0.006µs	nvfp4_dual_gemm.py
taka09203	69.694µs	+0.030µs	submission.py
jarvis_creates	69.739µs	+0.044µs	submission.py
roba5397	69.882µs	+0.143µs	submission.py

DizzleRama	69.966µs	+0.084µs	reference_soln_2.py
christian_6666	70.023µs	+0.056µs	submission.py
siro	70.024µs	+0.002µs	submission.py
hyunwon_	70.104µs	+0.080µs	submission.py
Figarland.S	70.124µs	+0.019µs	submission.py
Voldemort4321	70.281µs	+0.158µs	submission.py
Seraphim	71.949µs	+1.667µs	submission.py
joshisuyash	72.028µs	+0.079µs	submission_3.py
Evelyn A.	72.261µs	+0.234µs	submission.py
mklz	72.574µs	+0.312µs	reference.py
Ashley	74.091µs	+1.518µs	submission.py
listing_to_one_side	74.417µs	+0.325µs	submissionCC.py
AnDaltan	77.350µs	+2.934µs	submission.py
Her_77	81.998µs	+4.648µs	submission.py
pagelessuptime	82.166µs	+0.168µs	4_dual_gemm_submissic
Natalie Gollahon	82.305µs	+0.139µs	bs_improved.py
HohoHocCode	82.433µs	+0.128µs	submission.py
thuEcstasy	82.511µs	+0.078µs	ref.py
KFM89	82.551µs	+0.040µs	submission.py
c0de128	82.695µs	+0.144µs	submission.py
savik	85.483µs	+2.788µs	sub_v1.py
zyn	89.687µs	+4.204µs	submission.py
danielfrg	89.932µs	+0.245µs	test.py
Sherlock Holmes	90.049µs	+0.116µs	submission.py
misha_antonenko	108.377µs	+18.328µs	submission.py
ayghri	109.015µs	+0.639µs	submission.py
alice_hg22	109.114µs	+0.098µs	example.py
Julius T	115.134µs	+6.021µs	solution.py

Performance Trend

Contestant Search

guaguabear gau.nernst arseni_ivanov Simon macto

Type to search for contestants

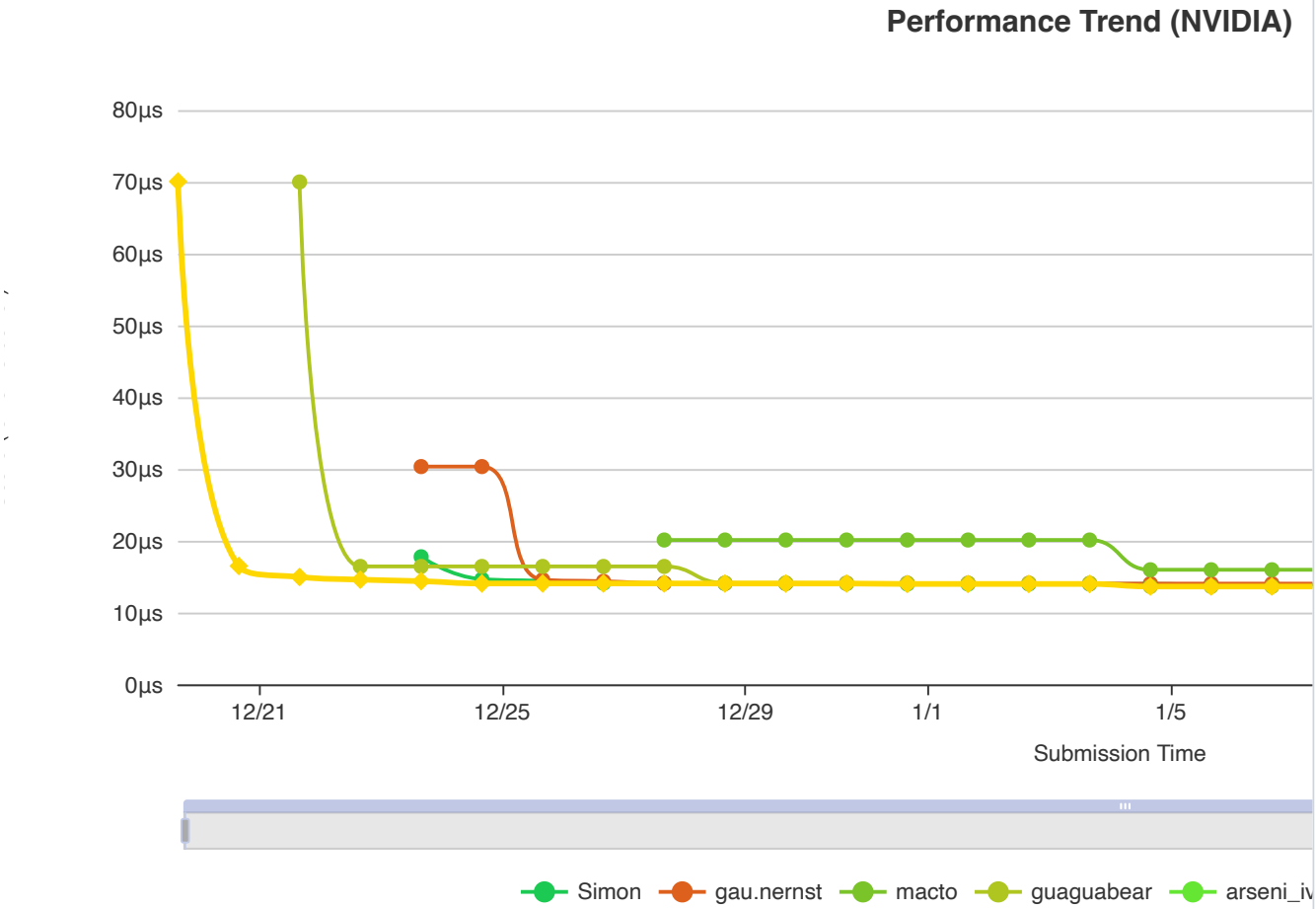
GPU Type

NVIDIA

Load Top 5

Best Over Time

Raw Submissions



X-Axis (Date): Start mm/dd/yyyy End mm/dd/yyyy

Y-Axis (µs): Min Max

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